

SHIELI, Kazakhstan

WMO: **380690**

Lat: **44.182N** Lon: **66.729E** Elev: **153** StdP: **99.50** Time Zone: **5.00 (E05)** Period: **01-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-21.0	-17.4	-24.1	0.4	-20.7	-20.9	0.6	-16.5	9.5	-10.2	8.4	-6.9	1.4	320	0.467	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	15.0	38.6	19.7	37.0	19.2	35.4	18.7	21.0	34.8	20.3	33.7	19.6	32.8	2.4	90	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	16.6	12.1	25.5	15.5	11.2	25.5	14.3	10.4	25.7	61.4	35.1	59.0	33.5	56.5	32.9	24.9	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	8.3	7.1	5.9	DB	-22.6	41.2	5.1	1.3	-26.3	42.1	-29.3	42.9	-32.2	43.7	-35.9	44.7	(4)
(5)				WB	-22.8	22.5	5.1	1.2	-26.5	23.3	-29.4	24.0	-32.3	24.7	-36.0	25.5	(5)

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	-3.8	-1.7	7.3	14.3	21.1	25.5	26.8	25.3	19.1	11.4	3.7	-2.7	(6)
(7)		DBStd	12.05	6.12	6.69	5.41	5.19	3.75	3.36	3.10	3.45	4.06	4.80	5.45	5.63	(7)
(8)		HDD10.0	1526	429	328	117	19	1	0	0	0	1	41	198	393	(8)
(9)		HDD18.3	3097	687	561	343	139	18	1	0	1	38	218	439	651	(9)
(10)		CDD10.0	2351	0	1	33	148	343	465	521	473	273	84	9	0	(10)
(11)		CDD18.3	882	0	0	1	18	102	216	263	216	62	3	0	0	(11)
(12)		CDH23.3	11898	0	0	21	260	1312	2954	3639	2801	829	81	1	0	(12)
(13)		CDH26.7	6283	0	0	4	79	561	1624	2111	1544	347	13	0	0	(13)
(14)	Wind	WSAvg	2.1	1.9	2.2	2.5	2.7	2.6	2.1	2.0	2.3	2.1	1.8	1.6	1.8	(14)
(15)	Precipitation	PrecAvg	148	18	15	16	19	21	8	5	3	3	9	18	14	(15)
(16)		PrecMax	277	48	49	40	86	84	30	41	18	18	34	65	41	(16)
(17)		PrecMin	67	3	1	2	2	1	0	0	0	0	0	1	2	(17)
(18)		PrecStd	51	11	10	9	16	20	9	9	4	4	9	15	9	(18)

				Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	17.3	26.0	32.3	35.8	40.1	41.4	39.7	35.7	28.5	21.8	11.6	(19)
(20)			MCWB	7.2	9.9	13.7	16.5	17.3	20.3	20.4	20.7	18.2	14.4	12.4	6.4	(20)
(21)		2%	DB	8.0	11.8	22.3	28.9	33.5	37.7	39.1	37.5	33.0	25.6	17.2	8.7	(21)
(22)			MCWB	4.8	7.4	12.0	15.2	16.7	19.3	19.9	19.4	16.9	13.1	10.5	5.2	(22)
(23)		5%	DB	5.4	9.1	19.0	26.2	31.8	36.0	37.2	35.8	30.8	23.1	14.4	6.8	(23)
(24)			MCWB	2.9	5.3	10.6	14.2	16.4	18.9	19.5	18.8	16.1	12.4	8.8	4.1	(24)
(25)		10%	DB	3.2	6.7	16.2	23.9	30.0	34.2	35.4	34.1	28.2	20.5	11.9	4.9	(25)
(26)			MCWB	1.3	3.6	9.6	13.3	15.9	18.2	19.1	18.2	15.0	11.1	7.6	2.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	10.4	14.6	17.8	19.2	21.7	22.2	22.0	18.8	15.6	13.9	7.6	(27)
(28)			MCDB	10.8	16.5	23.0	25.4	30.1	35.7	35.9	34.6	33.8	25.7	19.8	10.2	(28)
(29)		2%	WB	5.2	7.7	12.8	16.2	18.2	20.6	21.1	20.7	17.5	14.0	11.3	5.7	(29)
(30)			MCDB	7.7	11.2	20.7	25.9	29.6	34.2	34.7	34.6	31.2	23.9	16.6	8.0	(30)
(31)		5%	WB	3.1	5.6	11.3	15.1	17.3	19.7	20.4	19.7	16.5	12.7	9.4	4.2	(31)
(32)			MCDB	5.0	8.6	17.9	24.6	28.8	32.9	33.9	33.1	29.5	21.5	13.8	6.7	(32)
(33)		10%	WB	1.5	3.7	10.0	14.0	16.5	19.0	19.7	18.8	15.4	11.5	7.6	2.6	(33)
(34)			MCDB	3.2	6.7	15.8	23.1	27.8	32.1	33.0	32.0	27.3	19.9	11.6	4.9	(34)
(35)	Mean Daily Temperature Range		MDBR	8.4	9.8	11.9	12.5	14.0	14.8	15.0	15.3	15.3	14.1	10.7	8.9	(35)
(36)		5% DB	MCDBR	9.7	12.2	15.9	16.3	16.5	17.1	17.0	17.2	17.8	17.5	14.5	10.5	(36)
(37)			MCWBR	7.3	8.1	8.5	7.0	5.9	5.7	5.6	6.0	7.0	8.6	8.9	7.8	(37)
(38)		5% WB	MCDBR	9.0	11.5	14.0	14.2	14.3	15.4	15.3	15.9	16.9	16.1	12.6	9.2	(38)
(39)			MCWBR	7.0	7.9	8.1	6.6	5.8	5.8	5.6	6.3	7.0	8.4	8.0	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.356	0.384	0.440	0.479	0.478	0.460	0.446	0.418	0.388	0.400	0.387	0.357	(40)
(41)		taud		2.090	2.053	1.988	1.930	1.976	2.068	2.128	2.185	2.217	2.103	2.104	2.107	(41)
(42)		Ebn at Noon		720	774	780	785	800	816	823	834	829	746	676	673	(42)
(43)		Edh at Noon		103	128	156	179	176	161	150	137	122	120	101	92	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.50	2.60	3.97	5.47	6.97	7.62	7.48	6.76	5.32	3.42	1.96	1.33	(44)
(45)		RadStd		0.22	0.28	0.31	0.44	0.41	0.34	0.31	0.23	0.19	0.21	0.19	0.16	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-2.24	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+0.93	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page